

What are the costs and benefits of climate policy?

ECON-4210

Economics of Climate Change

Class 6

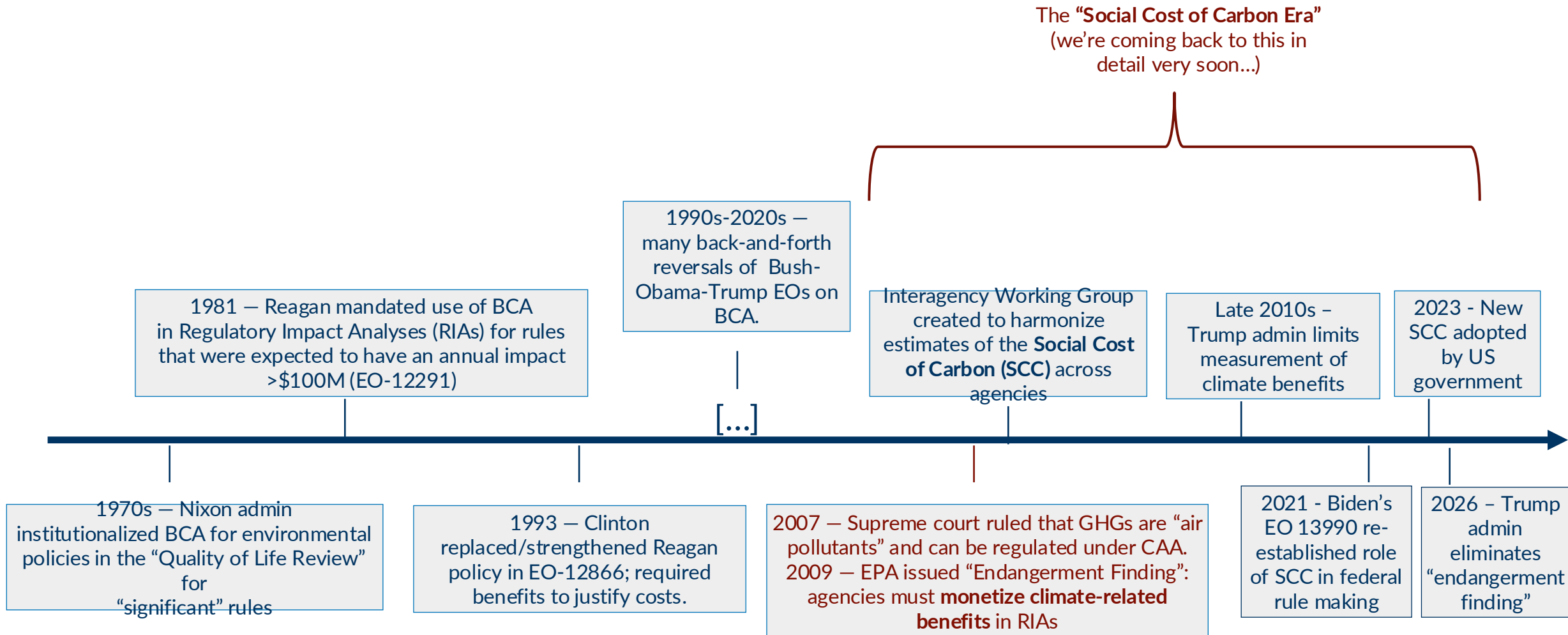
Prof. Casey Wichman

Today's class

1. Questions?
2. Attendance
3. Housekeeping
 1. First problem set due this week (due date: Friday, 9/18, 11:59 PM).
 2. Exam 1 is on Monday, 9/21.
4. **Analyzing costs and benefits of climate change**
5. For next class...

History of BCA and climate

Some key events in US regulatory history



The New York Times

Trump Administration

How Trump Stifles Speech

Voting by Mail

Video: Republican Support for Trump

Trump Administration Erases the Government's Power to Fight Climate Change

The Environmental Protection Agency rejected the bedrock scientific finding that greenhouse gases threaten human life and well being. It means the agency can no longer regulate them.

Listen · 12:59 min

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Rigorous scientific findings since 2009 have shown that greenhouse gases and global warming are harming public health. Jeff Swensen/Getty Images



By Lisa Friedman
Reporting from Washington

Feb. 12, 2026

<https://www.nytimes.com/2026/02/12/climate/trump-epa-greenhouse-gases-climate-change.html?searchResultPosition=3>

The New York Times

Trump Administration

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Voting by Mail

Video: Republican Support for Trump

E.P.A. Expected to Erase Limits on Climate Pollution From Power Plants

Generation of electricity is the second largest source of carbon dioxide and other planet-warming gases in the United States.

Listen · 7:22 min

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E.P.A. administrator Lee Zeldin speaking next to President Trump in the Oval Office earlier this year. Mr. Zeldin is expected to announce on Monday that the agency will no longer limit greenhouse gas emissions from power plants. Pete Marovich for The New York Times





By Maxine Joselow and Lisa Friedman
Reporting from Washington

Published Sept. 13, 2026 Updated Sept. 14, 2026, 10:34 a.m. ET

<https://www.nytimes.com/2026/09/13/climate/epa-power-plant-climate-rules.html?smid=nytcore-ios-share>

 Georgia Institute of Technology

What is benefit-cost analysis?

When should we use it, and why?

- Remember, economic efficiency = maximizing net benefits.
- So, maximizing benefits minus costs should be our goal in choosing among different policies.
 - A more lenient (and less “economic”) rule: if Benefits > Costs, do the project.
 - Or, among a set of potential policies, choose the policy that maximizes Benefits – Costs.
- In the US, any major regulation is required to perform a benefit-cost analysis to ensure that the benefits justify the costs (relative to alternative regulatory approaches)
 - **NB: you’ll see BCA and CBA used interchangeably. I like BCA simply because we want to do a Benefit-minus-Cost Analysis.

Steps in performing BCA

1. Set up the framework for analysis (define alternatives & baseline [i.e., “doing nothing”])
2. Decide whose costs & benefits should be recognized
3. Identify and categorize costs & benefits
4. Project costs and benefits over the life of each alternative
5. Monetize (place a dollar value on) costs and benefits
 - List benefits/costs that cannot be quantified
6. Compute net present value of benefits and costs
7. (Estimate distribution of costs and benefits to obtain expected present values)
8. Perform sensitivity analysis

Measuring climate benefits

Part 1

- **“With and without” principle**

- Only benefits that would result from the project should be counted
- Ignoring those that would have accrued anyway

- **Tangible vs. intangible benefits**

- Tangible benefits can be reasonably assigned a monetary value
- Intangible benefits cannot be assigned a monetary values
 - In some cases, “non-market valuation” techniques can be used to value intangible benefits

- **“Monetization” of benefits is key so that we have a common measuring stick.**

- This is a huge part of environmental/climate economics: putting a value on environmental damages.
- This is also where BCA gets attacked a lot.
- **Measuring benefits is difficult, but *not* including a benefit implicitly assumes that benefit is zero.**

Measuring climate benefits

Part 2

- We want to use willingness-to-pay (i.e., market demand) as a measure of benefits.
- For **market goods** and services this is easy.
 - Would you buy Prof. Wichman's iPad for \$50? What about \$1500? That tells me something about your WTP!
- For **nonmarket goods** (e.g., changes in environmental quality, or human health/mortality), this is more difficult.
 - Some things can be inferred via markets using **revealed preferences**
 - **Health risks / mortality** – often the largest "benefits" of environmental/climate policy is reducing mortality/morbidity.
 - **Hedonic methods** can be used, e.g., how much flood risk is capitalized into your home's value.
 - **Travel cost methods** can be used to determine WTP for recreational amenities (to value water quality improvements, reductions in wildfire risk)
 - Other benefits can be measure using **stated preference methods**
 - Structured surveys designed to elicit willingness to pay. Can be useful to inform **non-use values**
 - E.g., the value of polar bears existing, even if you never see one in the wild.

Estimating costs

- A cost is a real resource used up (e.g., tax revenue is a transfer, not a cost)
- **Measure by opportunity cost:**
 - What labor, capital, and fuel would have produced without the regulation?
- Two sources of cost estimates:
 - Ask the firms. Fast, but firms have every reason to overstate.
 - Model the engineering. No strategic bias, but engineering models miss how firms actually adjust — input switching, process changes, early retirement.
- Realized compliance costs for US air regulations have usually come in below the ex-ante estimates (Harrington, Morgenstern, and Nelson 2000).

Adding up costs and benefits over time

- Use a constant discount rate (e.g., 3%, 5%, 7%)
 - Costs and benefits accrue over 10, 20, 50 years or more...
 - The discount rate can be critical
 - We'll talk in much more detail about discounting for climate change after the exam...
- Calculate the **present value (PV)** of costs and benefits by adding up the discounted costs and benefits over time.
 - Choose a discount rate r and a time horizon T years.
 - Current value of benefits and costs over time are: $c_0, c_1, \dots, c_T$ and $b_0, b_1, \dots, b_T$.

$$\text{PV of benefits} = PV^{Benefits} = b_0 + \frac{b_1}{1+r} + \dots + \frac{b_T}{(1+r)^T} = \sum_{t=0}^T \frac{b_t}{(1+r)^t}$$

$$\text{PV of costs} = PV^{Costs} = c_0 + \frac{c_1}{1+r} + \dots + \frac{c_T}{(1+r)^T} = \sum_{t=0}^T \frac{c_t}{(1+r)^t}$$

- Net Present Value (NPV) = PV of Net Benefits = $NPV^{NetBenefits} = PV^{Benefits} - PV^{Costs}$

Where does the present value come from?

Imagine a stream of damages of \$1,000 per year for five years, discounted at $r=5\%$.

Year (t)	Damages	Discount factor $1/(1+r)^t$	Present value
0	\$1,000	1.0000	\$1,000
1	\$1,000	0.9524	\$952
2	\$1,000	0.9070	\$907
3	\$1,000	0.8638	\$864
4	\$1,000	0.8227	\$823
Total	\$5,000		\$4,546

\$5,000 of damages is worth \$4,546 today. At 2% the same stream is worth \$4,808.

How much does the discount rate matter?

Present value of **\$1 million** of climate damages occurring **50 years** from now.

Discount rate	Present value today
2%	\$371,500
3%	\$228,100
5%	\$87,200
7%	\$33,900

At 2% the present value is **11 times** the value at 7%.

BCA Criterion

- For a single project (vs. status quo):
 - → adopt the project if net present value (NPV) of the project's net benefits is positive.
- For multiple projects:
 - → select the project with the largest NPV (among mutually exclusive alternatives — with a fixed budget, rank by NPV per dollar spent).
- This BCA criterion is “**Potential Pareto Improvement (PPI)**” criterion (sometimes known as the **Kaldor-Hicks** criterion).
- A project satisfies the PPI criterion if:
 - Winners from a given policy *could* compensate the losers & still be better off
 - (doesn't require the compensation to actually take place!)
 - E.g., 45 of you give me \$1 each, and I give the \$45 plus \$10 of my own to one student → net benefits are +\$10, so BCA says do it — even though 44 of you are worse off.
 - The winner could hand back \$1 to each of them and still keep \$10.

Pitfalls with BCA

- Double counting
 - E.g., recreation value of new reservoir construction + increases in property value nearby reservoir
- Opportunity costs are real costs
 - What would have been the next best use of resources (e.g., land, real estate, capital)?
- Sunk costs
 - Just because some costs are irreversible doesn't mean you should necessarily follow through
- Taxes
 - Can create net social benefits (e.g., if used to correct market failures)
 - Otherwise, tax revenue is simply a transfer and irrelevant for CBA.
- Jobs...

Pitfalls with BCA

- Double counting
 - ▶ E.g., recreation value of new reservoir
- Opportunity costs are real costs
 - ▶ What would have been the next best use?
- Sunk costs
 - ▶ Just because some costs are irreversible
- Taxes
 - ▶ Can create net social benefits (e.g., if tax revenue is used for public goods)
 - ▶ Otherwise, tax revenue is simply a transfer
- Jobs...

Are jobs costs or benefits?

Yes!

“Under most circumstances, wages paid to people employed on a project should not be counted as benefits, although this practice is disturbingly widespread. As mentioned above, the opportunity cost of labor used on a project is the output that that labor would have generated elsewhere, either in paid work or in household production.”

-Zerbe and Bellas, “A Primer for Benefit-Cost Analysis” (2006), pg. 118

BCA in action with climate change...

- In 2007, the Supreme Court ruled that GHGs are air pollutants that could be regulated under the Clean Air Act if they affect health and welfare.
- In 2009, the EPA issued a report (the “Endangerment Finding”) that said GHGs affect health and welfare, which prompted new standards for reducing GHG emissions that states must achieve.
- In 2015, EPA issues rules to limit GHG from new power plants: **the Clean Power Plan (CPP)**
 - ▶ Goal: Achieve 32% reduction in GHG emissions by 2030 (relative to 2005 levels).
 - Sets state-specific CO2 goals on total emissions.
 - Sets maximum allowable CO2 emissions per unit of electricity for new power plants.
 - ▶ The CPP was never adopted as policy, but serves as a good example of BCA for climate change.

Clean Power Plan—Benefits

- Benefits calculated for all direct and ancillary (or co-benefits) of proposed rule.
- Climate benefits reflect global impacts
 - Social Cost of Carbon (SCC) is year specific and increases over time (\$17 to \$170 in 2030)
 - Discount rates considered: 2.5%, 3% and 5%
- Ancillary (or co-) benefits that are monetized are health impacts of SO₂, NO_x and PM_{2.5} reductions of 25% by 2030.

Benefits Category	Specific Effect	Effect Has Been Quantified	Effect Has Been Monetized	More Information
Improved Environment				
Reduced climate effects	Global climate impacts from CO ₂	— ¹	✓	SC-CO ₂ TSD
	Climate impacts from ozone and black carbon (directly emitted PM)	—	—	Ozone ISA, PM ISA ²
	Other climate impacts (e.g., other GHGs such as methane, aerosols, other impacts)	—	—	IPCC ²
Improved Human Health (co-benefits)				
Reduced incidence of premature mortality from exposure to PM _{2.5}	Adult premature mortality based on cohort study estimates and expert elicitation estimates (age >25 or age >30)	✓	✓	PM ISA
	Infant mortality (age <1)	✓	✓	PM ISA
Reduced incidence of morbidity from exposure to PM _{2.5}	Non-fatal heart attacks (age > 18)	✓	✓	PM ISA
	Hospital admissions—respiratory (all ages)	✓	✓	PM ISA
	Hospital admissions—cardiovascular (age >20)	✓	✓	PM ISA
	Emergency room visits for asthma (all ages)	✓	✓	PM ISA
	Acute bronchitis (age 8-12)	✓	✓	PM ISA
	Lower respiratory symptoms (age 7-14)	✓	✓	PM ISA
	Upper respiratory symptoms (asthmatics age 9-11)	✓	✓	PM ISA
	Asthma exacerbation (asthmatics age 6-18)	✓	✓	PM ISA
	Lost work days (age 18-65)	✓	✓	PM ISA
	Minor restricted-activity days (age 18-65)	✓	✓	PM ISA
	Chronic Bronchitis (age >26)	—	—	PM ISA ²
	Emergency room visits for cardiovascular effects (all ages)	—	—	PM ISA ²
	Strokes and cerebrovascular disease (age 50-79)	—	—	PM ISA ²
	Other cardiovascular effects (e.g., other ages)	—	—	PM ISA ³
	Other respiratory effects (e.g., pulmonary function, non-asthma ER visits, non-bronchitis chronic diseases, other ages and populations)	—	—	PM ISA ³
	Reproductive and developmental effects (e.g., low birth weight, pre-term births, etc)	—	—	PM ISA ^{3,4}
	Cancer, mutagenicity, and genotoxicity effects	—	—	PM ISA ^{3,4}
Reduced incidence of mortality from exposure to ozone	Premature mortality based on short-term study estimates (all ages)	✓	✓	Ozone ISA
	Premature mortality based on long-term study estimates (age 30–99)	—	—	Ozone ISA ²
Reduced vegetation and ecosystem effects from exposure to ozone	Visible foliar injury on vegetation	—	—	Ozone ISA ²
	Reduced vegetation growth and reproduction	—	—	Ozone ISA ²
	Yield and quality of commercial forest products and crops	—	—	Ozone ISA ²
	Damage to urban ornamental plants	—	—	Ozone ISA ³
	Carbon sequestration in terrestrial ecosystems	—	—	Ozone ISA ²
	Recreational demand associated with forest aesthetics	—	—	Ozone ISA ³
	Other non-use effects	—	—	Ozone ISA ³
	Ecosystem functions (e.g., water cycling, biogeochemical cycles, net primary productivity, leaf-gas exchange, community composition)	—	—	Ozone ISA ³
Reduced effects from acid deposition	Recreational fishing	—	—	NO _x SO _x ISA ²
	Tree mortality and decline	—	—	NO _x SO _x ISA ³
	Commercial fishing and forestry effects	—	—	NO _x SO _x ISA ³
	Recreational demand in terrestrial and aquatic ecosystems	—	—	NO _x SO _x ISA ³
	Other non-use effects	—	—	NO _x SO _x ISA ³
	Ecosystem functions (e.g., biogeochemical cycles)	—	—	NO _x SO _x ISA ³

Clean Power Plan—Benefits

- In 2030, with 3% discount rate on climate benefits
 - ▶ **Total GHG benefits: \$20B**
 - ▶ Adding in co-benefits: **\$31-\$48B**
 - Depending on discount rate for health co-benefits
 - Health co-benefits arise from reduced PM2.5 and Ozone pollution (driven primarily by VSL!)
 - ▶ Climate only at 5%: **\$6.4B**
 - ▶ Climate only at 2.5%: **\$29B**
 - ▶ **Discount rates can influence climate benefits dramatically...**

Table ES-8. Combined Estimates of Climate Benefits and Health Co-benefits for Mass-Based Approach (billions of 2011\$)*

SC-CO ₂ Discount Rate and Statistic**	Climate Benefits Only	Climate Benefits plus Health Co-benefits (Discount Rate Applied to Health Co-benefits)					
		3%			7%		
In 2020	82	million short tons CO ₂					
5%	\$0.94	\$2.9	to	\$5.7	\$2.8	to	\$5.3
3%	\$3.3	\$5.3	to	\$8.1	\$5.1	to	\$7.7
2.5%	\$4.9	\$6.9	to	\$9.7	\$6.7	to	\$9.3
3% (95 th percentile)	\$9.7	\$12	to	\$14	\$11	to	\$14
In 2025	264	million short tons CO ₂					
5%	\$3.6	\$11	to	\$21	\$10	to	\$19
3%	\$12	\$19	to	\$29	\$18	to	\$27
2.5%	\$17	\$24	to	\$35	\$24	to	\$33
3% (95 th percentile)	\$35	\$42	to	\$52	\$42	to	\$51
In 2030	413	million short tons CO ₂					
5%	\$6.4	\$18	to	\$34	\$17	to	\$32
3%	\$20	\$32	to	\$48	\$31	to	\$46
2.5%	\$29	\$41	to	\$57	\$40	to	\$55
3% (95 th percentile)	\$60	\$72	to	\$89	\$71	to	\$86

*All benefit estimates are rounded to two significant figures. Climate benefits are based on reductions in CO₂

Clean Power Plan—Benefits & Costs

- Compliance costs = costs at state level associated with meeting emissions reductions.
- Compliance costs also include monitoring and enforcement.
- States are free to reduce emissions by various means
 - E.g., less coal, more renewables (and NG), cap-and-trade markets, energy efficiency programs, etc.
- **NPV of net benefits in 2030:**
 - **\$25-\$45B**
- *Note that “climate-only” benefits at 5% are smaller than costs...*

Table ES-9. Monetized Benefits, Compliance Costs, and Net Benefits Under the Rate-based Illustrative Plan Approach (billions of 2011\$) ^a

	Rate-Based Approach					
	2020		2025		2030	
Climate Benefits ^b						
5% discount rate	\$0.80		\$3.1		\$6.4	
3% discount rate	\$2.8		\$10		\$20	
2.5% discount rate	\$4.1		\$15		\$29	
95th percentile at 3% discount rate	\$8.2		\$31		\$61	
	Air Quality Co-benefits Discount Rate					
	3%	7%	3%	7%	3%	7%
Air Quality Health Co-benefits ^c	\$0.70 to \$1.8	\$0.64 to \$1.7	\$7.4 to \$18	\$6.7 to \$16	\$14 to \$34	\$13 to \$31
Compliance Costs ^d	\$2.5		\$1.0		\$8.4	
Net Benefits ^e	\$1.0 to \$2.1	\$1.0 to \$2.0	\$17 to \$27	\$16 to \$25	\$26 to \$45	\$25 to \$43

Non-monetized climate benefits

Reductions in exposure to ambient NO₂ and SO₂

Reductions in mercury deposition

Ecosystem benefits associated with reductions in emissions of NO_x, SO₂, PM, and mercury

Visibility impairment

Non-Monetized Benefits

Residual thoughts on BCA

- BCA can help to formalize decision-making, but often is not (nor should be) the only consideration.
 - Helps to choose among alternatives
 - Helps to identify and reject inefficient policies
 - Does not guarantee economic efficiency (but real-world policy is messy and efficiency is rarely, if ever, achieved)
- Fairness also matters
 - Discounting future benefits/costs necessarily “discounts” the welfare of future generations for problems like climate change.
 - Benefits/costs accrue to different people today, and the value of an additional \$1 to a relatively poor person is much greater than the value of an additional \$1 to Jeff Bezos.

For next class...

<https://www.caseyjwichman.com/econ4210/#week4>